

Anojan Yogenthiran

+1 (226)-456-1210 | ayogenthiran.ai@gmail.com | LinkedIn | GitHub | Portfolio | Medium

SKILLS

Languages: Python, SQL (PostgreSQL), TypeScript

GenAI/LLMs: RAG, LLM Fine-Tuning, Agentic AI, MCP, Prompt Engineering, LLM Evaluation, Guardrails, HITL Workflows

AI/ML: Predictive Modeling, NLP, Deep Learning, Transformers, Feature Engineering, Model Evaluation, Classification

Frameworks: PyTorch, Scikit-learn, Hugging Face, LangChain, LangGraph, LlamaIndex, LangSmith, FastAPI, vLLM

MLOps/Cloud: AWS, Azure, Docker, Kubernetes, MLflow, Optuna, OpenEvals, GitHub Actions, GitLab CI/CD

WORK EXPERIENCE

Machine Learning Associate

Jan 2026 – Present

Vector Institute

Toronto, Ontario

- Designed a conversational ReAct scheduling agent using LangGraph, LangChain, and Azure OpenAI to capture physician availability, preferences, and constraints in natural language, reducing scheduling intake time by **90%**.
- Built a real-time validation layer using prompt engineering and structured tool orchestration to ground agent decisions in scheduling data, improving physician constraint-handling reliability.
- Led an agent evaluation pipeline with LangSmith, OpenEvals, and AgentEvals to benchmark trajectory accuracy and tool-call correctness, catching **15+** failure modes before production.
- Implemented PII redaction, prompt-injection defenses, I/O schema filtering, and HITL validation to prevent sensitive data leakage in production.
- Deployed the scheduling agent on Kubernetes via GitLab CI/CD and Azure Container Registry, automating production delivery.

Applied AI Developer

May 2025 – Dec 2025

Transpots

Toronto, Ontario

- Architected a document automation system using LangChain, LangGraph, and GPT-4o to extract and validate factoring documents, reducing manual processing time by **80%**.
- Built a GPT-4 validation pipeline with rule-based guardrails and HITL checks, achieving **95%+** extraction accuracy and reducing false approvals by **35%**.
- Deployed agentic document workflows as FastAPI microservices for production factoring workflows, enabling automated extraction, validation, and approval testing.

Data Science and Engineering Intern

Jan 2025 – Apr 2025

Kaidu.ai

Toronto, Ontario

- Built a Bluetooth RSSI-based motion classification model using DBSCAN and Dynamic Time Warping, achieving **89%** accuracy for stationary vs. mobile detection.
- Reduced false positives by **83%** by integrating Dynamic Fourier smoothing into the Bluetooth RSSI preprocessing pipeline, improving motion-detection reliability.
- Selected optimal production model via MLflow experiment tracking and Optuna hyperparameter tuning, improving inference stability by **25%**.
- Presented deep learning feasibility analysis to the CEO, evaluating model viability, implementation cost, and ROI for production motion analytics.

Research Associate

Apr 2024 – Dec 2024

Intelligence Data Science Lab, Western University

London, Ontario

- Fine-tuned LLaMA-3.2 and Mistral-7B using QLoRA for miRNA-target prediction, improving domain-specific generation accuracy by **15%** over base models.
- Optimized DeepSELEX CNN for DNA-binding site prediction using hyperparameter tuning and custom preprocessing, improving F1-score by **20%**.
- Automated HT-SELEX genomic data pipelines with Airflow, enabling scalable ingestion and reproducible experiment tracking.

PROJECTS

ArxivLens — AI Research Paper Assistant | [GitHub](#)

- Built a RAG research assistant over **3M+** ArXiv papers using ChromaDB, embeddings, FastAPI, and Streamlit, delivering source-cited Q&A with metadata filtering, cross-encoder reranking, retrieval evaluation, and **90%+** retrieval accuracy.

Telco Churn Prediction — Customer Retention ML System | [GitHub](#)

- Deployed an Optuna-tuned XGBoost churn prediction system on AWS ECS Fargate with FastAPI serving, MLflow tracking, and CI/CD, achieving **92%+** recall through an end-to-end containerized inference pipeline.

EDUCATION

Western University

London, Ontario

M.Sc. in Computer Science, Specialization in Artificial Intelligence; GPA: 3.9/4.0

Sep 2023 – Feb 2025

- Research:** Generative AI-Driven Aptamer Discovery for Accelerated Drug Development
- Relevant Coursework:** Statistical Machine Learning, Advanced AI, Reinforcement Learning, Advanced Databases

Sri Lanka Institute of Information Technology

Colombo, Sri Lanka

B.Sc. (Honours) in Electrical and Electronic Engineering

Jan 2018 – Dec 2021